

## Data Monitoring Platform Brief

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### Technical Requirements Brief: Multi-Tenant Data Monitoring Platform Overview

The Client needs an agile product developer to orchestrate an AI-driven build platform (such as Base44 or a similar full-stack AI engine) to rapidly construct a functional Data Capture Loop and Dual-Data Routing Architecture. The system must seamlessly integrate a digital public channel (MailChimp/Typeform) and an offline-first mobile field channel. By leveraging the native velocity of an AI development engine, the goal is to bypass traditional technical debt and deploy a live, interactive ecosystem that routes clean, compliant, and structured data into isolated client and analytical data views.

#### Phase 1 Execution: Modular Dashboard Configuration & Data Target Mapping

The developer will configure the initial data schema relationships directly within the AI engine, mapping out a multi-tenant, white-label presentation terminal. To ensure maximum resilience during stakeholder consultations, the interface must adhere to a modular, variable-driven "Defensive Design" strategy:

##### 1. Reusable Component Containers (The Page 1 Metric Grid)

- **The Blueprint:** The six headline cross-portfolio metrics on Page 1 must be built as identical, decoupled layout slots.
- **The Component:** Each container must be bound to exactly three generic variables: `metric_title`, `metric_value`, and a traffic-light `compliance_state` indicator.
- **The Purpose:** This allows instant, zero-code field reassignment if a client flags a metric as non-essential, swapping data vectors without breaking the global visual layout.

##### 2. Category-Level Data Abstraction (Pages 2–4 Detailed Capture Feeds)

- **The Blueprint:** Data tables rendering granular records across the typology deep-dive screens must abstract specific question text out of the UI column headers.
- **The Component:** Grid columns for the generalized impact variables (\$Q4\$ through \$Q9\$) will be labeled strictly by broad thematic categories (e.g., "*Spatial Transit Sentiment (Q4)*") instead of exact question strings.
- **The Purpose:** This decouples the presentation frontend from text-level changes made to survey inputs post-pitch.

##### 3. Core Architectural Pillars

The wireframe schema must treat the following data pipelines as permanent, structurally fixed system inputs:

- **User Contextual Filters (Q1–Q3):** Demographic brackets, asset class selectors, and tenure matrices that drive the interface filtering logic.
- **Qualitative NLP Engine (Q10):** A dedicated, hard-capped 280-character text box or voice transcription field, paired with a color-coded backend sentiment tag indicator.

## Summary of Upcoming Phases

In Phase 2, the developer will use the AI platform to configure the data capture interfaces, establishing a zero-keyboard mobile PWA wrapper with local browser caching ([IndexedDB](#)) for field teams alongside automated URL parameter tracking for outbound digital surveys. Phase 3 will focus on backend execution, utilizing the platform's internal database layer to wire an event-driven split controller. This controller will securely isolate granular tenant data via Row-Level Security in Stream A while irreversibly anonymizing data points into a macro data pool for Stream B. Full contract continuation is contingent on providing comprehensive repository documentation detailing the architecture layout, API connections, and synchronization rules within a clear [README.md](#) file.

[Phase 2: Data Capture Layouts] >> Tailors Input Fields to Match Schema Perfectly

[Phase 3: Backend Execution] >> Codes the Pipeline & Dual-Routing Split